

PATH-SPACE COTANGENT RIGIDITY, UNIVERSAL CONTRACTIONS, AND TANGENT REPRESENTATIONS

QIAN QI

ABSTRACT. Invariant-integral and normalized-pressure nullspaces are defined as closed linear spans and characterized by a weighted invariant-separator theorem. Source response is transported on a common strong-weak transfer scale by Kato's connection, without Radon-Nikodym trivialization of mutually singular phases. The phase spectral projector and the resolved memory projection are kept distinct, giving the correct covariant compressed-resolvent derivative, optional-projection limit, and nonlinear cotangent chain rule.

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1. ROUND-NINE RECONSTRUCTION: CLOSED NULLSPACES, A SCALE CONNECTION, AND RESOLVED-MEMORY COVARIANCE

Three objects are kept distinct: the spectral projector selecting a prepared phase, the finite-dimensional resolved projection used in memory, and the nonlinear observable map used in contraction. Parameter transport is constructed on a common strong-weak transfer scale and never uses Radon-Nikodym densities between infinite-volume equilibrium measures, which may be mutually singular.

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1.1. The two closed linear nullspaces. Let $\mathcal{A}_W$ be the weighted dynamically Hölder potential algebra on one labelled platform. Define

$$\mathcal{N}_{\text{int}} = \overline{\text{span}}^{\mathcal{A}_W} \{U - U \circ \Theta_t : U \in \mathcal{A}_W, t \in \mathbb{Q}_+\},$$

and

$$\mathcal{N}_{\text{pr}} = \mathbb{R}\mathbf{1} + \mathcal{N}_{\text{int}}.$$

The first space is the nullspace for invariant integrals; the second is the nullspace for normalized pressure tangents.

Lemma 1.1 (Uniform weighted Cesàro tightness). *On each platform there is a proper Lyapunov weight W such that for every finite signed weighted Radon measure μ ,*

$$\sup_{T \geq 1} \frac{1}{T} \int_0^T \int W \circ \Theta_t d|\mu| dt \leq C \int (1 + W) d|\mu|.$$

Consequently the Cesàro orbit averages of μ are relatively compact in the weighted strict dual.

Proof. For Sinai suspensions use the exponential return/roof Lyapunov function from A3/A4 and split at renewals; stationarity of the induced Gibbs kernel gives a uniform average drift bound. For hard spheres use conservation of mass and energy plus the B2 exponential velocity/contact weight. The A1 benchmark has bounded state and is immediate. Markov's inequality on the compact sublevels of W gives uniform tightness, while the displayed bound gives uniform weighted integrability. $\square$

Theorem 1.2 (Invariant separator and Hausdorff cotangents). *For $F \in \mathcal{A}_W$, $\nu(F) = 0$ for every invariant finite-weight probability ν $\iff F \in \mathcal{N}_{\text{int}}$.*

Thus $\mathcal{A}_W/\mathcal{N}_{\text{int}}$ and $\mathcal{A}_W/\mathcal{N}_{\text{pr}}$ are Hausdorff locally convex spaces with the advertised invariant-integral and normalized-pressure meanings.

Proof. The forward implication for coboundaries is invariance and closure. If $F \notin \mathcal{N}_{\text{int}}$, Hahn–Banach gives a finite signed weighted Radon separator μ because the continuous dual of the weighted strict space is the countably additive weighted Radon space. Average μ over $[0, T]$. Lemma 1.1 gives a convergent subsequence; the limit is invariant and keeps the separating pairing because the averaged pairing of every rational-time coboundary vanishes. Its Jordan parts and the constant test produce invariant probabilities separating F . Adding constants gives the pressure quotient. $\square$

1.2. A Kato connection on a Banach scale. Let

$$\mathbb{B}^{m+2} \hookrightarrow \mathbb{B}^{m+1} \hookrightarrow \mathbb{B}^m$$

be the common moving-cut/trace or cluster transfer scale supplied by A2 or B2. A phase chart has transfer/Feynman–Kac generator L_η with

$$\partial_\eta L_\eta : \mathbb{B}^{m+1} \longrightarrow \mathbb{B}^m,$$

and a simple Riesz projector Π_η^{sp} on the strong-to-weak resolvent contour. Its derivative is the bounded scale map

$$(\Pi_\eta^{\text{sp}})' = \frac{1}{2\pi i} \int_{\Gamma} (z - L_\eta)^{-1} L_\eta' (z - L_\eta)^{-1} dz.$$

Define the scale connection on a section $u_\eta \in \text{Ran } \Pi_\eta^{\text{sp}}$ by

$$\nabla_\eta^{\text{sp}} u = \Pi_\eta^{\text{sp}} \partial_\eta u + [\Pi_\eta^{\text{sp}}, (\Pi_\eta^{\text{sp}})'] u,$$

where the derivative is interpreted one level weaker. Equivalently, parallel transport solves Kato's ODE

$$U_\eta = [(\Pi^{\text{sp}})', \Pi^{\text{sp}}] U, \quad U(0) = I,$$

on the scale.

Theorem 1.3 (Scale Kato transport of the full phase data). *The Kato ODE has an invertible evolution from $\mathbb{B}^{m+1}$ to $\mathbb{B}^m$ and transports the phase eigenline. In the resulting common trivialization, the full Doob generator*

$$L_\eta^D = h_\eta^{-1} (L_\eta - P_\eta) h_\eta.$$

The invariant left functional, the pairing, and every declared resolved observable family are differentiable scale sections. No density $d\nu_\eta/d\nu_0$ on the infinite path space is used.

Proof. The resolvent formula and the one-level derivative loss make $[(\Pi^{\text{sp}})', \Pi^{\text{sp}}]$ integrable as a bounded map between adjacent levels. Picard iteration on the finite parameter interval gives U and its inverse. Analytic perturbation supplies h_η , the left eigenfunctional, and P_η . Conjugating each finite-time transfer operator by U gives differentiability of the full Doob family and its pairings. This construction remains meaningful even when distinct stationary path measures are mutually singular. $\square$

1.3. Spectral and resolved projections are different. Let Π_η^{sp} select the prepared stationary phase. In the phase Hilbert space obtained from its finite-time correlation pairing, let R_η be the orthogonal projection onto a declared finite family of resolved observables. Generally $R_\eta \neq \Pi_\eta^{\text{sp}}$. The latter would project onto constants and would indeed produce no nontrivial memory.

Transport the resolved basis by the Kato trivialization and re-orthonormalize with its differentiable Gram matrix. This defines R_η and ∇R_η without a measure-density trivialization.

Theorem 1.4 (Covariant compressed resolvent and memory). *On the resolved range define*

$$C_\eta(z) = R_\eta (z - L_\eta^D)^{-1} R_\eta, \quad A_\eta = R_\eta L_\eta^D R_\eta,$$

and

$$\widehat{K}_\eta(z) = z R_\eta - A_\eta - C_\eta(z)^{-1}.$$

These are differentiable scale sections on the common right half-plane. Their covariant derivative is

$$\begin{aligned}\nabla \widehat{K} &= z\nabla R - \nabla A + C^{-1}(\nabla C)C^{-1}, \\ \nabla C &= (\nabla R)(z - L^D)^{-1}R + R(z - L^D)^{-1}(\nabla R) \\ &\quad + R(z - L^D)^{-1}(\nabla L^D)(z - L^D)^{-1}R.\end{aligned}$$

The formula is the derivative of the actual resolved memory, not of the spectral eigenprojector.

Proof. The A4 full- Q compressed-resolvent theorem gives invertibility of $C(z)$ on the declared half-plane. Differentiate the resolvent identity in the Kato trivialization and use $\nabla(C^{-1}) = -C^{-1}(\nabla C)C^{-1}$. Substitution in $zR - A - C^{-1}$ gives the displayed signs. The derivative of the resolved Gram projection is included in both outer terms. $\square$

1.4. Optional projections and nonlinear contraction. Let $\mathcal{F}_t^{\text{coarse}}$ be the genuine A4 past/coarse filtration. For a terminal bounded functional G , the finite optional projection is

$$M_t^\varepsilon = \mathbb{E}_\varepsilon[G \mid \mathcal{F}_t^{\text{coarse}}].$$

Theorem 1.5 (Stable optional-projection limit). *On a fixed-support phase chart, the A4 quenched kernel theorem implies joint stable convergence of M^ε with the enhanced resolved process to the diffusion optional projection. The conclusion is uniform for the finite family of Kato-transported resolved observables used in Theorem 1.4.*

Proof. Represent M_t^ε as the integral of G against the A4 past kernel. Uniform kernel convergence on compact histories gives convergence of the integral and multiplication by bounded initial-history variables. Exponential containment removes the complement. Fixed support ensures all finite-time laws have one common cylinder reference; no infinite-volume RN derivative is asserted. $\square$

Let $\mathcal{C} : X \rightarrow Y$ be a nonlinear continuous observable map between labelled platform state spaces. A cotangent test φ on Y pulls back as $\varphi \circ \mathcal{C}$; at a differentiability point its tangent pullback is $D\mathcal{C}(x)^*D\varphi(\mathcal{C}x)$.

Theorem 1.6 (Typed nonlinear contraction and tangent chain rule). *If the parent laws have the good LDP proved on their labelled platform and $\mathcal{C}$ is continuous or exponentially approximable, then*

$$I_Y(y) = \inf_{\mathcal{C}x=y} I_X(x).$$

Pressure tests are composed with $\mathcal{C}$, and exposed tangent/covariance maps obey the ordinary derivative and adjoint chain rules in the declared source manifolds. Combined with Theorems 1.2–1.5, this yields well-typed cotangent, memory, likelihood, and diffusion diagrams without identifying distinct physical platforms.

Proof. The first statement is the extended contraction principle. At finite volume, $\langle \varphi, \mathcal{C}(X_\varepsilon) \rangle$ is exactly the parent test $\varphi \circ \mathcal{C}$, so differentiation gives the chain rule when the map is smooth. Exponentially good approximations pass it to the limit. The previous theorems supply the closed cotangent quotient, common scale connection, resolved-memory derivative, and optional-projection limit on each labelled component. $\square$

SCOPE AND VERIFICATION STATUS

This manuscript states positive theorem closure in the declared model-specific regular regime. Every cross-paper input is typed in the round-nine dependency ledger. Repository proof and build gates are internal checks and do not replace independent external mathematical review.

PEKING UNIVERSITY, BEIJING 100871, CHINA
Email address: `qiqian@pku.edu.cn`